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Recent progress in borate-based short-wavelength nonlinear optical crystals with boron–oxygen skeleton modification

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Ultraviolet (UV, $\lambda < 400$ nm) and deep-UV ($\lambda < 200$ nm) nonlinear optical (NLO) crystals have the ability to convert the wavelength of laser to output UV or even deep-UV coherent light by means of frequency doubling and play a significant role in solid-state lasers. Recently, borate-based short-wavelength NLO crystals have attracted considerable attention owing to their diverse structures and tunable performance. For example, fluorooxoborates, hydroxyborates, hydroxyfluorooxyborates, borophosphates, borosilicates, borosulfates and fluoroborophosphates display rich structure types and enhanced optical properties. In this review, recently discovered borate-based NLO crystals with a modified boron–oxygen skeleton are overviewed and analyzed, including their crystal structure features, short-wave cutoff edges, birefringence and second harmonic generation (SHG) responses. This review presents the positive influence of boron–oxygen skeleton modification on the structure–property relationship. We hope to provide the recent progress in the crystal chemistry of borate-based short-wavelength NLO crystals and facilitate the discovery of promising UV and even deep-UV NLO candidates.

1. Introduction

Second-order nonlinear optical (NLO) crystal materials are widely used in many optoelectronic fields (laser communication, laser machining, industrial processing, medical diagnosis, *etc.*) owing to their ability to generate tunable coherent light through frequency conversion technology, and the design and synthesis of NLO crystal materials in the short wavelength ultraviolet (UV) spectral range are the research frontier in this field.[1–6] Since the first nonlinear optical experiment of second harmonic generation (SHG) using quartz crystal was carried out in 1961,[7] nonlinear optics and NLO materials have developed rapidly. A variety of NLO crystals have been found, including borates, carbonates, nitrates, sulfates, phosphates, silicates, *etc.*[8–18] Meanwhile, borate is a preferred system for studying short-wavelength UV NLO crystals owing to its rich structural chemistry and excellent properties.[19–21] Many commercially available borate NLO crystals, such as $KBe_2BO_3F_2$ (KBBF),[22] β-BaB_2O_4 (BBO),[23] LiB_3O_5 (LBO)[24] and $CsLiB_6O_{10}$ (CLBO),[25] are used to meet the application requirements of short-wavelength UV region owing to their performance advantages (including wide transparency range (band gap), large SHG coefficient, phase matching (birefringence) and the ease of obtaining large-size crystals). However, there is no perfect NLO crystal that can meet large-scale applications. KBBF can directly output deep-UV laser through frequency conversion, but its layered growth habit and the toxicity of raw material BeO limit its commercial development.[26] The study of borate-based short-wavelength UV NLO crystals is still in progress.

The microstructure of the crystal plays a decisive role in its macroscopic properties.[27–29] In the process of exploring the relationship between the structure and properties of the NLO crystals, Chen systematically studied the relationship between the SHG coefficient and the microstructure of different types of oxide crystals, and proposed the anionic group theory.[30] The core ideas of the theory include the following: (1) the NLO-active functional units are the anionic groups; (2) in the first-order approximation, the main group cations have little effect; (3) the macroscopic frequency doubling coefficient corresponds to the geometric superposition of the microscopic frequency doubling coefficient of the anionic groups. In summary, to design and construct short-wavelength borate-based UV NLO crystals with excellent performance, the selection and optimal arrangement of the dominant functional units in the borate anionic skeleton is important. Based on the theory, the chemical modification of the borate anionic skeleton is an important strategy to obtain new borate-based NLO crystals with excellent properties.[31,32] A typical modification method is

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anionic derivatization, such as the formation of fluorooxoborates, hydroxyborates and hydroxyfluorooxoborates.[33–35] Another common modification method is to incorporate non-metal P(v), Si, and S(vi) atoms into the skeleton to form the so-called borophosphates, borosilicates and borosulfates.[36–39] Except for the above modification method, partial metal atoms (Be, Zn and Al) can form covalent tetrahedra that are similar to $[BO_4]$, $[PO_4]$, $[SiO_4]$ and $[SO_4]$ units and improve the optical properties of crystals in the borate system.[40–42] In 2021, Pan *et al.* reviewed the influence of the covalent tetrahedra $[MO_{4-n}X_n]$ (n = 1–3) on the optical properties.[51] The orbitals of B and O atoms can be sp or sp^2 hybridized to form $[BO_2]$ and $[BO_3]$ units, or sp^3 hybridized to form $[BO_4]$ units.[43,44] For anionic derivatization, (1) F atoms with large electronegativity and the extranuclear electron arrangement of $2s^22p^5$ are introduced into the $[BO_4]$ units of borates, and the $[BO_xF_{4-x}]$ (x = 1–3) units have larger polarizability anisotropy and hyperpolarizability than BO_4 units; (2) the terminal O atom of $[BO_3]$ and $[BO_4]$ units can often combine with the H atom to form $[BO_x(OH)_{3-x}]$ (x = 0–2), $[BO_2(OH)_2]$ and $[B(OH)_4]$ units, the introduction of H atoms eliminates the dangling bond of terminal O to increase the energy gap, and on the other hand, hydrogen bonds will optimize the arrangement of anionic units; (3) the synergistic introduction of F and H atoms will combine their advantages. By incorporating non-metal P(v), Si and S(vi) atoms into the borate anionic skeleton, tetrahedral $[BO_4]$ units will condense with $[PO_4]$, $[SiO_4]$ and $[SO_4]$ units by B–O–S, B–O–Si, B–O–P bonding. Although $[PO_4]$, $[SiO_4]$ and $[SO_4]$ units have small hyperpolarizability and polarizability anisotropy, they have large energy gaps and the P(v), Si and S (vi) atoms may bring unexpected results to borates. In order to produce UV or even deep-UV output, the shortest phase-matching wavelength (λ_{PM}) of the material in question needs to be at least less than 400 nm or 200 nm. The shortest phase-matching wavelength of crystals is determined by the UV cutoff edge and refractive index dispersion (in general, the shorter the cutoff edge, the smaller the change in refractive index, and the larger the birefringence, the shorter the shortest phase matching wavelength(λ_{PM})).[45,46] For short-wavelength UV NLO crystals, it is easier to obtain a phase-matching wavelength shorter than 400 nm or even 200 nm, by combining the advantages of π and non-π conjugated units in the borate anionic skeleton.

In recent years, many borate-based short-wavelength UV NLO crystals, in which the shortest phase-matching wavelengths are less than 266 nm or even partially less than 200 nm, are discovered. For example, much attention has been attracted by fluorooxoborates, hydroxyborates and hydroxyfluorooxyborates with anionic derivatization, such as $MgB_5O_7F_3$,[52] $NaSr_3(OH)(B_9O_{16})[B(OH)_4]$[65] and $[C(NH_2)_3][B_3O_3F_2(OH)_2]$.[67] Moreover, fluoroborophosphates with a derivatized $[BO_3F]$ unit and $[PO_4]$ unit have been reported as DUV NLO materials. In this paper, we will systematically summarize the recent research progress of borate-based short-wavelength UV NLO crystals with various borate anionic skeletons. Their stoichiometric composition, anionic units, space groups, cutoff edges, birefringence, SHG responses and shortest phase-matching wavelengths are summarized in Table 1. For the convenience of discussion, they are divided into three categories according to different chemical modification methods of the borate anionic skeleton: (1) anionic derivatization; (2) boron–oxygen skeleton modification with $[MO_4]$ (M = P, Si and S) units; (3) synergistic modification of derivatized $[BO_3F]$/$[BO_2(OH)]$ units and the $[PO_4]$ unit. Finally, the future development opportunities and challenges of borate-based short-wavelength NLO crystals based on chemical modification are briefly discussed.

2. Short-wavelength NLO crystals by anionic derivatization

In this section, the results of borate anionic derivatization involve three types of fluorooxoborates, hydroxyborates, and hydroxyfluorooxyborates. The relationship between the crystal structures and nonlinear optical properties of the above crystals and their nonlinear reactive motifs is also discussed in detail.

2.1. Fluorooxoborates

In 2017, Pan's group proposed the design strategy of replacing the O atom in $[BO_4]$ with a F atom and introducing the formed $[BO_xF_{4-x}]$ (x = 1, 2, and 3) units into the borate anionic skeleton, which helps to reduce the structural symmetry while maintaining the intrinsic advantage of excellent nonlinear optical properties of borates, which will help to obtain noncentrosymmetric (NCS) structures.[33,47–51] In this section, newly developing fluorooxoborates are discussed.

$MgB_5O_7F_3$[52] crystallizes in the space group $Cmc2_1$ with the orthorhombic system, which features a *zigzag* $^2[B_5O_7F_3]_\infty$ 2D-layered crystal structure composed by $[BO_3]$ and $[BO_3F]$ units, with Mg cations filling the intra-layers. $MgB_5O_7F_3$ exhibits a strong SHG response of approximately 2.4 × KDP, and the calculated birefringence is 0.07@1064 nm. Moreover, the available experimental data show that it has a UV cut-off edge below 200 nm and the shortest type I SHG phase-matching wavelength is down to ~189 nm.

$MgB_5O_7F_3$, $CaB_5O_7F_3$[53] and $SrB_5O_7F_3$[54,55] crystallized in the same $Cmc2_1$ space group and they are isostructural, we chose the newly discovered $MgB_5O_7F_3$ for further discussion. As shown in Fig. 1, the FBB of $MgB_5O_7F_3$ is the naphthalene-like ring $[B_5O_9F_3]$, which can be obtained from two $[BO_3]$ units and three $[BO_3F]$ units by sharing O atoms. The FBBs of $[B_5O_9F_3]$ are further connected to each other by sharing O atoms to form *zigzag* $^2[B_5O_7F_3]_\infty$ layers, which are stacked along the [010] direction to form the final anionic skeleton with Mg cations filling the intra-layers. At the same time, the dangling bond of the O atom in $[BO_3]$ is eliminated by the attached $[BO_3F]$, so that all three isostructural compounds have UV cut-off edges shorter than 200 nm. And they exhibit relatively strong SHG responses of about 2.4, 2.0 and 1.6 × KDP. The shortest phase-matching wavelengths of $MgB_5O_7F_3$, $CaB_5O_7F_3$ and $SrB_5O_7F_3$ are evaluated to be 189, 180 and 183 nm, respectively, indicating that it has potential to produce deep-UV coherent light by a direct SHG process.

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Table 1 Optical properties of selected borate-based NLO crystals

Compounds	Anionic units	Space groups	Cutoff edges	Birefringence (@1064 nm)[b]	SHG response[a] (@1064 nm)	λ_{PM}	Ref.
$Na_4B_8O_9F_{10}$	$[BO_3]$, $[BO_2F_2]$, $[BF_4]$	$Pna2_1$	<200 nm	0.036	0.7 × KDP	240 nm	58
$MgB_5O_7F_3$	$[BO_3]$, $[BO_3F]$	$Cmc2_1$	<200 nm	0.07	2.4 × KDP	189 nm	52
$CaB_5O_7F_3$	$[BO_3]$, $[BO_3F]$	$Cmc2_1$	<180 nm	0.07	2 × KDP	180 nm	53
$SrB_5O_7F_3$	$[BO_3]$, $[BO_3F]$	$Cmc2_1$	<180 nm	0.07	1.6 × KDP	183 nm	54
$LiNaB_6O_9F_2$	$[BO_4]$, $[BO_2F_2]$, $[BO_3]$	$Pnn2$	<200 nm	0.067	1.1 × KDP	210 nm	56
$NaRb_6(B_4O_5(OH)_4)_3(BO_2)$	$[BO_2]$, $[BO_2(OH)]$, $[BO_3(OH)]$	$R32$	<200 nm	0.044	0.1 × SiO_2	N/A	59
$CaB_8O_{11}(OH)_4$	$[BO_3]$, $[BO_4]$, $[BO(OH)_2]$, $[BO_2(OH)]$	$P2_1$	<200 nm	0.089	2 × KDP	174 nm	60
$SrB_8O_{11}(OH)_4$	$[BO_3]$, $[BO_4]$, $[BO(OH)_2]$, $[BO_2(OH)]$	$P2_1$	<200 nm	0.083	1.8 × KDP	185 nm	60
$NaSr_3(OH)(B_9O_{16})[B(OH)_4]$	$[B(OH)_4]$, $[BO_3]$, $[BO_4]$	$P3_1c$	<190 nm	0.039	3.3 × KDP or 0.55 × β-BBO @ 532 nm	≤266 nm	65
$LiBa_3(OH)(B_9O_{16})[B(OH)_4]$	$[B(OH)_4]$, $[BO_3]$, $[BO_4]$	$P3_1c$	<200 nm	0.016	1.2 × KDP or 0.33 × β-BBO @ 532 nm	≤266 nm	66
$[C(NH_2)_3][B_3O_3F_2(OH)_2]$	$[BO_2(OH)]$, $[BO_2F_2]$	$P1$	~200 nm	0.17	0.9 × KDP or 0.3 × β-BBO @ 532 nm	≤266 nm	67
$[C(NH_2)_3]_2[B_3O_3F_4(OH)]$)	$[BO_2(OH)]$, $[BO_2F_2]$	$P1$	~200 nm	0.16	1.4 × KDP or 0.5 × β-BBO @ 532 nm	≤266 nm	67
$Rb_3B_{11}P_2O_{23}$	$[BO_3]$, $[BO_4]$, $[PO_4]$	$P1$	168 nm	0.07	2.5 × KDP	205 nm	69
$Rb_3B_8PO_{16}$	$[BO_3]$, $[BO_4]$, $[PO_4]$	Cc	175 nm	0.07	0.5 × KDP	222 nm	70
$Cs_2B_4SiO_9$	$[BO_4]$, $[SiO_4]$	$I\bar{4}$	<190 nm	N/A	4.6 × KDP	N/A	71
$CsSiB_3O_7$	$[BO_3]$, $[BO_4]$, $[SiO_4]$	$Pna2_1$	<200 nm	0.035[c]	0.8 × KDP	N/A	72
$(NH_4)_2B_4SO_{10}$	$[BO_3]$, $[BO_4]$, $[SO_4]$	$C2$	184 nm	0.053	1.1 × KDP	252 nm	73
$K_2B_4SO_{10}$	$[BO_3]$, $[BO_4]$, $[SO_4]$	$C2$	182 nm	0.057	0.9 × KDP	228 nm	74
$Rb_2B_4SO_{10}$	$[BO_3]$, $[BO_4]$, $[SO_4]$	$C2$	190 nm	0.059	0.9 × KDP	239 nm	74
$Cs_2B_4SO_{10}$	$[BO_3]$, $[BO_4]$, $[SO_4]$	$C2$	189 nm	0.06	1.7 × KDP	245 nm	74
$(NH_4)_3B_{11}PO_{19}F_3$	$[BO_3]$, $[BO_4]$, $[BO_3F]$, $[PO_4]$	$R3$	183 nm	0.088	1.2 × KDP	190 nm	77
$Rb_3B_{11}PO_{19}F_3$	$[BO_3]$, $[BO_4]$, $[BO_3F]$, $[PO_4]$	$R3$	<200 nm	0.08	1.3 × KDP	180 nm	74
$K_2B_4PO_9(OH)$	$[BO_3]$, $[BO_2(OH)]$, $[BO_4]$, $[PO_4]$	$P2_1$	<200 nm	0.055	N/A	224 nm	70

[a] Powder SHG efficiency was recorded using the Kurtz and Perry method;[78] N/A = not available or not reported. [b] Calculated birefringence by the DFT method. [c] The wavelength has not been reported.

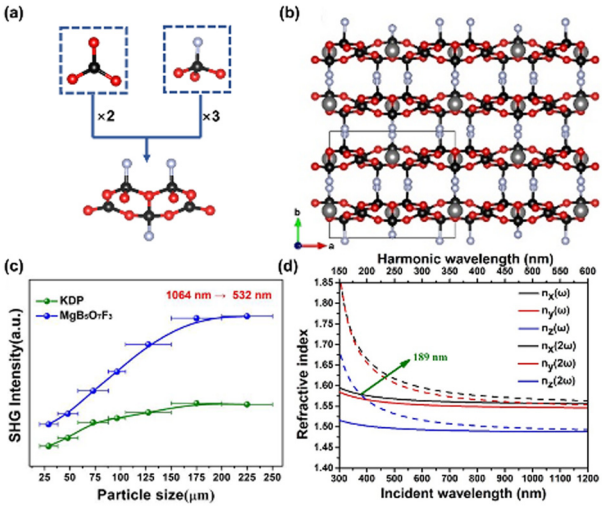

Fig. 1 (a) Anionic units of the $[B_5O_9F_3]$ FBB; (b) the crystal structure of $MgB_5O_7F_3$; (c) the powder SHG data for $MgB_5O_7F_3$ under 1064 nm laser radiation; (d) refractive index dispersion curves for fundamental and second-harmonic light of $MgB_5O_7F_3$.[45] Parts c and d are reproduced with permission. Copyright 2021 Wiley.

$LiNaB_6O_9F_2$[56] crystallizes in the orthorhombic space group $Pnn2$ and $Li_2B_6O_9F_2$[57] crystallizes in the monoclinic space group Cc. They were synthesized by a solid state reaction method in the closed system. For $LiNaB_6O_9F_2$, it features two independent chains of 1D $[B_6O_{10}F_2]$ forming the 3D framework of novel DNA-like double helixes with Li/Na cations residing between chains to balance the charge (Fig. 2). The FBB of $LiNaB_6O_9F_2$ is $[B_6O_{11}F_2]$ consisting of one $[BO_4]$, one $[BO_2F_2]$ and four $[BO_3]$ units. $LiNaB_6O_9F_2$ possesses a deep-UV cutoff edge, large SHG response (1.1 × KDP), moderate birefringence (0.067 at 1064 nm) and shortest phase-matching wavelength (210 nm). The FBB of $Li_2B_6O_9F_2$ is $[B_6O_{11}F_2]$ and its crystal structure consists of a $[B_6O_9F_2]$ 3D framework. Similar to $LiNaB_6O_9F_2$, it shows a deep-UV cutoff edge (below 190 nm), a comparable SHG response of 0.9 × KDP, and large birefringence (0.07@1064 nm).

$Na_4B_8O_9F_{10}$[58] crystallizes in the space group $Pna2_1$ of the orthorhombic crystal system, which features 1D zigzag $[B_7O_9F_6]$

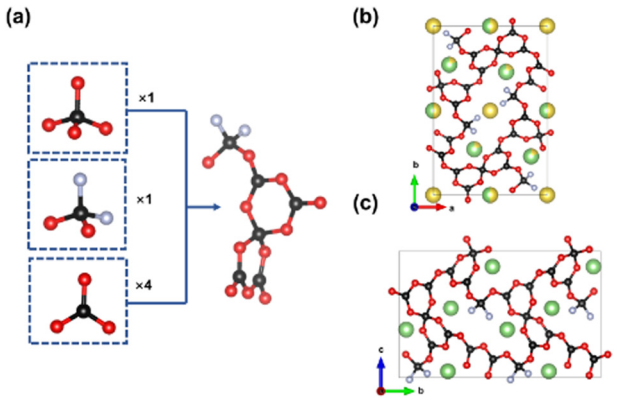

Fig. 2 (a) Anionic units of $[B_6O_{11}F_2]$; (b) the crystal structure of $LiNaB_6O_9F_2$; (c) the crystal structure of $Li_2B_6O_9F_2$.

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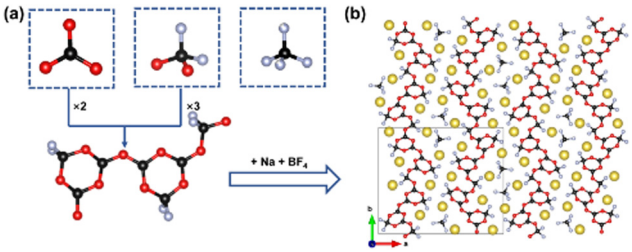

Fig. 3 (a) Anionic units of $[B_7O_{10}F_6]$ and $[BF_4]$; (b) the crystal structure of $Na_4B_8O_9F_{10}$.

chains with isolated $[BF_4]$ tetrahedra and Na cations residing between chains to balance the charge. As shown in Fig. 3, the FBB of $Na_4B_8O_9F_{10}$ is composed of $[B_7O_{10}F_6]$ and $[BF_4]$, where four $[BO_3]$ and three $[BO_2F_2]$ form the $[B_7O_{10}F_6]$ unit by sharing O atoms, and the $[B_7O_{10}F_6]$ unit is further connected to form a 1D $[B_7O_9F_6]$ chain. Optical measurements reveal that $Na_4B_8O_9F_{10}$ exhibits a comparable SHG response of ~0.7 × KDP, and a short UV cut-off edge below 200 nm. In addition, the shortest phase-matching wavelength is 240 nm with a small birefringence (0.036@1064 nm). All these properties clearly demonstrate that $Na_4B_8O_9F_{10}$ is an important NLO crystal for UV applications.

2.2. Hydroxyborates

Hydroxyborates are similar to fluorooxoborates that the hydroxyl group is directly bonded to the B atom to form a hydroxyborate, which may have similar effects to fluorination: enriching the structure of the borate and improving the properties of the compound. In this section, newly developing hydroxyborates are discussed.

$NaRb_6(B_4O_5(OH)_4)_3(BO_2)$[59] with the *R*32 space group, an NCS and chiral mixed-alkali-metal borate with a linear $[BO_2]$ unit, was synthesized by a solvothermal method. The $[B_4O_5(OH)_4]$ and $[BO_2]$ units construct the borate anionic skeleton, where the two $[BO_2(OH)]$ and two $[BO_3(OH)]$ are connected by sharing O atoms to form 0 D $[B_4O_5(OH)_4]$ clusters (Fig. 4). And Rb or Na cations are located in the structure to balance the charge. $NaRb_6(B_4O_5(OH)_4)_3(BO_2)$ shows a deep-UV cutoff edge (200 nm), a moderate birefringence (0.044@1064 nm) and small SHG response (0.1 × SiO_2). Although $NaRb_6(B_4O_5(OH)_4)_3(BO_2)$ has two NLO-active units ($[BO_3(OH)]$ and $[BO_2]$) in the structure, powder SHG measurements

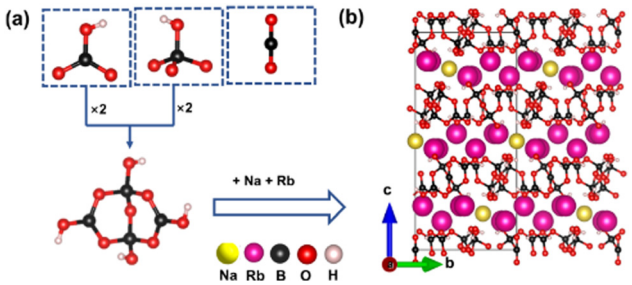

Fig. 4 (a) Anionic units of $[B_4O_5(OH)_4]$ and $[BO_2]$; (b) the crystal structure of $NaRb_6(B_4O_5(OH)_4)_3(BO_2)$.

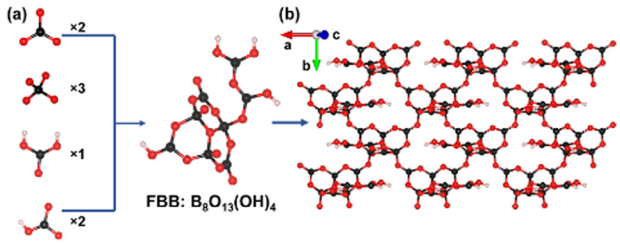

Fig. 5 (a) Anionic units of $[B_8O_{13}(OH)_4]$; (b) the 2D $^2[B_8O_{11}(OH)_4]_\infty$ layers.

show a small SHG response. One of the reasons is that measured samples of $NaRb_6(B_4O_5(OH)_4)_3(BO_2)$ exhibit two different enantiomers (the right-handed enantiomer and the left-handed enantiomer).

In 2021, the linear/nonlinear optical properties of $CaB_8O_{11}(OH)_4$ and $SrB_8O_{11}(OH)_4$[60] were systematically investigated by Lin's team based on first-principles calculations. $CaB_8O_{11}(OH)_4$[61–63] and $SrB_8O_{11}(OH)_4$[64] crystallize in the same space group $P2_1$ (No. 4); we chose $SrB_8O_{11}(OH)_4$ for further discussion. As shown in Fig. 5, the FBB of $SrB_8O_{11}(OH)_4$ is an assembly of two $[BO_3]$, three $[BO_4]$, one $[BO(OH)_2]$, and two $[BO_2(OH)]$ units to obtain the complex $[B_8O_{13}(OH)_4]$, while the $[B_8O_{13}(OH)_4]$ FBBs are interconnected with each other by the shared oxygen atoms to form 2D $^2[B_8O_{11}(OH)_4]_\infty$ layers, and these layers are stacked along the *c*-direction to form the final 3D anionic framework, with Sr cations filling between the layers to balance the charge. For $SrB_8O_{11}(OH)_4$, its terminal O atom is connected to the H atom, eliminating the hanging bond of the O atom, so the compound achieves a blue shift of the cut-off edge and deep UV transmission. Also, $CaB_8O_{11}(OH)_4$ and $SrB_8O_{11}(OH)_4$ have large SHG responses (2.0 × and 1.8 × KDP) and large birefringence (0.089 and 0.083@1064 nm). Theoretical calculations show that $CaB_8O_{11}(OH)_4$ and $SrB_8O_{11}(OH)_4$ are potential deep-UV NLO crystals with the shortest phase-matching wavelengths of 174 and 185 nm, respectively.

$NaSr_3(OH)(B_9O_{16})[B(OH)_4]$[65] and $LiBa_3(OH)(B_9O_{16})[B(OH)_4]$[66] crystallize in the $P3_1c$ space group and they are isostructural; we chose $NaSr_3(OH)(B_9O_{16})[B(OH)_4]$ for further discussion. As shown in Fig. 6, the FBB of $NaSr_3(OH)(B_9O_{16})[B(OH)_4]$ is composed of $[B_9O_{19}]$ and $[B(OH)_4]$, where three $[BO_3]$ and six $[BO_4]$ form the $[B_9O_{19}]$ unit by sharing O atoms, and the $[B_9O_{19}]$ unit is further connected to form a 3D framework with six-membered ring channels along the *c*-axis. $NaSr_3(OH)(B_9O_{16})[B(OH)_4]$ and $LiBa_3(OH)(B_9O_{16})[B(OH)_4]$ have large SHG responses (3.3 × and 1.2 × KDP), deep-UV cutoff edge (below 200 and 190 nm) and small birefringence (0.016 and 0.039@1064 nm).

2.3. Hydroxyfluorooxyborates

Two examples of organic-inorganic hybrid hydroxyfluorooxyborates ($[C(NH_2)_3][B_3O_3F_2(OH)_2]$ and $[C(NH_2)_3]_2[B_3O_3F_4(OH)]$)[67] with *P*1 space groups were designed and synthesized based on $[BO_2(OH)]$ and $[BO_2F_2]$ units. The structure of $[C(NH_2)_3][B_3O_3F_2(OH)_2]$ consists of π-conjugated $[C(NH_2)_3]$ and $[BO_2(OH)]$ and non-π-conjugated $[BO_2F_2]$ units (Fig. 7). The overall structure of $[C(NH_2)_3][B_3O_3F_2(OH)_2]$ can be regarded as a stacking of $^2\{[C(NH_2)_3][B_3O_3F_2(OH)_2]\}_\infty$ layers

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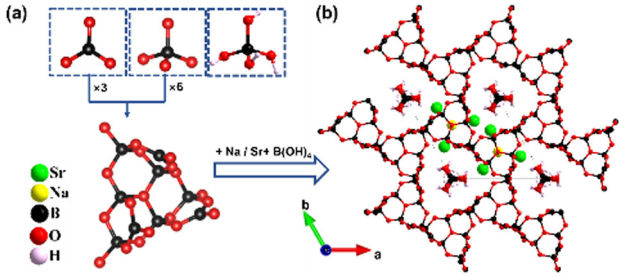

Fig. 6 (a) Anionic units of $[B_9O_{19}]$ and $[B(OH)_4]$; (b) the crystal structure of $NaSr_3(OH)(B_9O_{16})[B(OH)_4]$.

consisting of $[C(NH_2)_3]$ and $[B_3O_3F_2(OH)_2]$ units along a specific direction. The overall structure of $[C(NH_2)_3]_2[B_3O_3F_4(OH)]$ consists of $^2\{[C(NH_2)_3][B_3O_3F_4(OH)][C(NH_2)_3]\}_\infty$ two-dimensional layers assembled by $[C(NH_2)_3]$ and $[B_3O_3F_4(OH)]$ heteroanionic units along a specific orientation. One $[BO_2(OH)]$ and two $[BO_2F_2]$ units trimerize with each other by sharing a bridging O atom to form the $[B_3O_3F_4(OH)]$ heteroanion. Both examples of guanidine hydroxyfluorooxyborates are characterized by layered structures with NLO activity and exhibit excellent linear and nonlinear optical properties, including short UV cutoff edges (~200 nm), large SHG response (0.9–1.4 × KDP @ 1064 nm) and moderate birefringence (0.17 and 0.16@1064 nm), which are potential short-wavelength UV NLO crystals.

3. Boron–oxygen skeleton modification with $[MO_4]$ (M = P, Si and S) units

3.1. Borophosphates

The classic borophosphate BPO_4[68] with asymmetric $[BO_4]$ and $[PO_4]$ units possesses a large SHG response (2 × KDP), wide UV transmission range (130 nm) and small birefringence (0.005@1064 nm). However, small birefringence is not conducive to birefringent phase matching. The strategy of incorporating P(v) atoms in the borate skeleton with $[BO_3]$ and $[BO_4]$ units to form a B-O-P linked anionic skeleton helps to reduce the structural symmetry

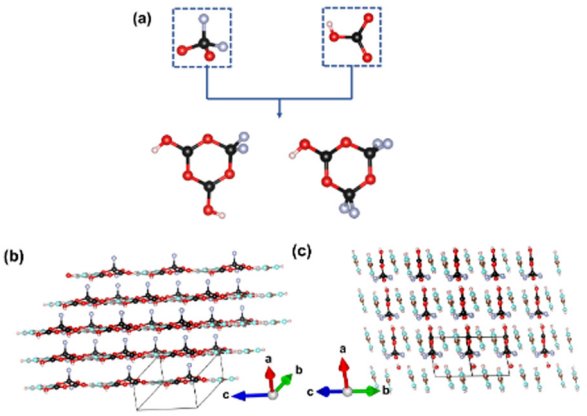

Fig. 7 (a) Anionic units of $[B_3O_3F_2(OH)_2]$ and $[B_3O_3F_4(OH)]$; (b) the crystal structure of $[C(NH_2)_3][B_3O_3F_2(OH)_2]$; (c) the crystal structure of $[C(NH_2)_3]_2[B_3O_3F_4(OH)]$.

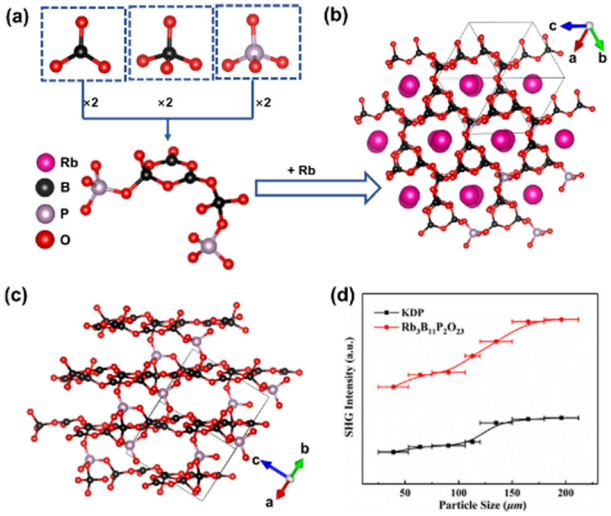

Fig. 8 (a) Anionic units of the $[B_4P_2O_{16}]$ FBB; (b) the crystal structure of $Rb_3B_{11}P_2O_{23}$; (c) the 3D $[B_{11}P_2O_{23}]_\infty$ framework; (d) the powder SHG data for $Rb_3B_{11}P_2O_{23}$ under 1064 nm laser radiation.[62] Part d is reproduced with permission. Copyright {2023} American Chemical Society.

and increase the birefringence while maintaining the large bandgap.

$Rb_3B_{11}P_2O_{23}$[69] crystallizes in the NCS *P*1 space group. As shown in Fig. 8, the FBB of $Rb_3B_{11}P_2O_{23}$ is composed of $[B_4P_2O_{16}]$, where two $[BO_3]$, two $[BO_4]$ and two $[PO_4]$ units form the $[B_4P_2O_{16}]$ unit by sharing O atoms, and the $[B_4P_2O_{16}]$ unit is further connected by B–O–B and B–O–P to form a 3D $[B_{11}P_2O_{23}]_\infty$ framework with an 18-membered ring. The Rb cations fill the $[B_{11}P_2O_{23}]_\infty$ framework to balance the charges. $Rb_3B_{11}P_2O_{23}$ has a large SHG response (2.5 × KDP), a deep-UV cutoff edge (168 nm), and a large birefringence (0.07@1064 nm) and its shortest phase-matching wavelength is 205 nm.

$Rb_3B_8PO_{16}$[70] belongs to the *Cc* space group. The FBB of $Rb_3B_8PO_{16}$ is composed of $[B_8PO_{20}]$, where five $[BO_3]$, three $[BO_4]$ and one $[PO_4]$ units form the $[B_8PO_{20}]$ unit by sharing O atoms, and the $[B_8PO_{20}]$ unit is further connected by B–O–B and B–O–P to form a 2D $[B_8PO_{16}]_\infty$ double layer with Rb cations residing in the intra-/inter-layers to balance the charge (Fig. 9). $Rb_3B_8PO_{16}$ has a small SHG response (0.5 × KDP), and it may arise from the almost antiparallel arrangement of the NLO-active motifs $[BO_3]$. The coexistence of $[BO_3]$, $[BO_4]$ and $[PO_4]$ units causes the $Rb_3B_8PO_{16}$'s deep-UV cutoff edge (175 nm), large birefringence (0.07@1064 nm) and its shortest phase-matching wavelength of 222 nm.

3.2. Borosilicates

$Cs_2B_4SiO_9$[71] and $CsSiB_3O_7$[72] crystallize in the space groups $I\bar{4}$ and $Pna2_1$. The FBB of $Cs_2B_4SiO_9$ is composed of $[B_4SiO_{13}]$, where four $[BO_4]$ and one $[SiO_4]$ unit form $[B_4SiO_{13}]$ units by sharing O atoms, and the $[B_4SiO_{13}]$ units are further connected by B–O–B and B–O–Si to form a 3D framework. Unlike the fully tetrahedral $[B_4SiO_{13}]$ FBB, the $[B_3SiO_{10}]$ FBB of $CsSiB_3O_7$ is composed of two planar $[BO_3]$, one $[BO_4]$ and one $[SiO_4]$ unit. Similar to $[B_4SiO_{13}]$ units, the $[B_3SiO_{10}]$ units are further

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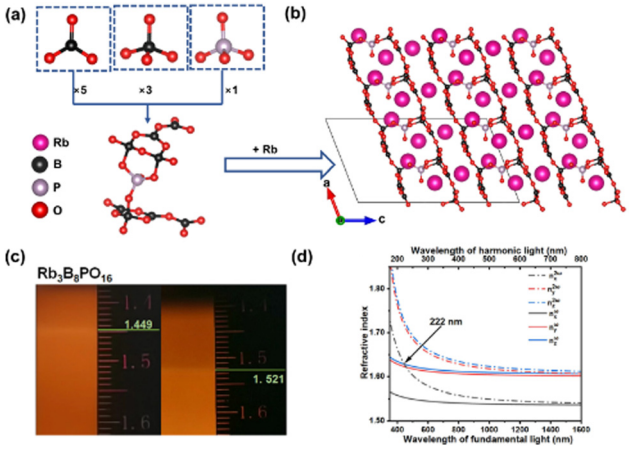

Fig. 9 (a) Anionic units of the $[B_8PO_{20}]$ FBB; (b) the crystal structure of $Rb_3B_8PO_{16}$; (c) photographs showing the measured n_{max} and n_{min} of the $Rb_3B_8PO_{16}$ crystal on the refractometer; (d) refractive index dispersion curves for fundamental and second-harmonic light of $Rb_3B_8PO_{16}$.[63] Parts c and d are reproduced with permission. Copyright {2022} American Chemical Society.

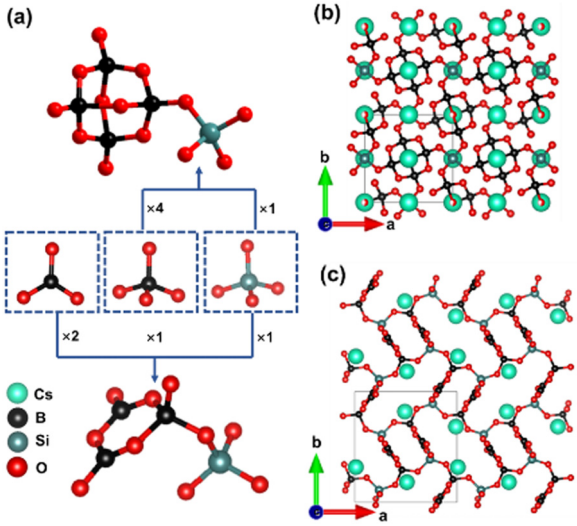

Fig. 10 (a) Anionic units of $[B_4SiO_{13}]$ (top) and $[B_3SiO_{10}]$ (bottom) FBBs; (b) the crystal structure of $Cs_2B_4SiO_9$ in the *ab* plane; (b) the crystal structure of $CsSiB_3O_7$ in the *ab* plane.

connected by B–O–B and B–O–Si to form a 3D framework (Fig. 10). $Cs_2B_4SiO_9$ and $CsSiB_3O_7$ possess relatively large SHG responses (4.6 × KDP and 0.8 × KDP) and deep-UV cutoff edges (below 190 nm and below 200 nm) in inorganic metal borosilicate systems. The calculated birefringence of $CsSiB_3O_7$ is 0.035.

3.3. Borosulfates

The discovery of a series of boron-rich borosulfates ($M_2B_4SO_{10}$ (M = NH_4, K, Rb and Cs)) with $[BO_3]$, $[BO_4]$ and $[SO_4]$ units NLO crystals has recently injected fresh blood into UV NLO crystals, which have novel $[B_4SO_{10}]_\infty$ layered structure and significant enhancement in birefringence compared to borosulfates containing only rigid tetrahedral motifs.

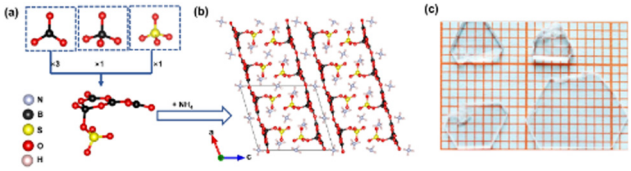

Fig. 11 (a) Anionic units of the $[B_4SO_{12}]$ FBB; (b) the crystal structure of $(NH_4)_2B_4SO_{10}$; (c) as-grown crystal for $(NH_4)_2B_4SO_{10}$.[66] Part c is reproduced with permission. Copyright 2021 Wiley.

In the $M_2B_4SO_{10}$ family, the crystal structures are very similar. $M_2B_4SO_{10}$ (M = NH_4, K, Rb and Cs)[73,74] all crystallize in the space group *C*2 and they are isotropic. As shown in Fig. 11, the three $[BO_3]$, one $[BO_4]$ and one $[SO_4]$ units are connected by sharing O atoms to form $[B_4SO_{12}]$ FBBs and FBBs are further bonded by sharing O atoms to form 2D infinite $[B_4SO_{10}]_\infty$ layers, with cations filling between layers to balance the charge. The $[BO_4]$ units eliminate the dangling bond of terminal oxygen of $[BO_3]$ units and the UV transparency can be guaranteed. All $M_2B_4SO_{10}$ (M = NH_4, K, Rb and Cs) have excellent NLO performance. The UV cutoff edges are 184, 182, 190 and 189 nm and the SHG responses are 1.1 ×, 0.9 ×, 0.9 × and 1.7 × KDP for $(NH_4)_2B_4SO_{10}$, $K_2B_4SO_{10}$, $Rb_2B_4SO_{10}$ and $Cs_2B_4SO_{10}$, respectively. The results of theoretical calculation show that the $[B_4SO_{10}]_\infty$ layers of the $M_2B_4SO_{10}$ family are the main origin for the SHG responses. In addition, the parallel layered structure causes a large optical anisotropy and they all have a large birefringence. The calculated birefringence values at 1064 nm are 0.053, 0.0.057, 0.0.059, and 0.060 for $(NH_4)_2B_4SO_{10}$, $K_2B_4SO_{10}$, $Rb_2B_4SO_{10}$ and $Cs_2B_4SO_{10}$, respectively. The moderate birefringence is beneficial to achieve UV phase-matching. For example, the shortest type-I phase-matching wavelengths for $(NH_4)_2B_4SO_{10}$, $K_2B_4SO_{10}$, $Rb_2B_4SO_{10}$ and $Cs_2B_4SO_{10}$ are 252, 228, 239 and 245 nm, respectively, Thus, these crystals are the candidates for UV frequency conversions.

4. Synergistic modification of derivatized $[BO_3F]/[BO_2(OH)]$ units and $[PO_4]$ unit

The earlier reported $ABPO_4F$ (A = K, Rb, Cs, and NH_4)[75,76] series and $(NH_4)_2BPO_4F_2$ of fluoroborophosphates have short UV cutoff edges (below 200 nm) and 0.3–1.1 × KDP SHG response; however, for the $ABPO_4F$ (A = Rb, Cs, and NH_4) series and $(NH_4)_2BPO_4F_2$, they have small (0.004@1064 nm for $(NH_4)_2BPO_4F_2$) or zero ($RbBPO_4F$, $CsBPO_4F$ and NH_4BPO_4F) birefringence and they are all phase mismatched. Only $KBPO_4F$ with the calculated birefringence of 0.044@1064 nm is proved that it satisfies the phase matching condition at 1064 nm. In the recently reported fluoroborophosphates, enhanced birefringence was achieved by the coexistence of $[BO_3]$, $[BO_4]$, $[BO_3F]$ and $[PO_4]$ units in the anionic skeleton.

$M_3B_{11}PO_{19}F_3$ (M = NH_4 and Rb)[74,77] crystallize in the same space group *R*3, and we chose $Rb_3B_{11}PO_{19}F_3$ for further discussion. Three $[BO_3]$, one $[BO_4]$, one $[BO_3F]$ and one $[PO_4]$ units are

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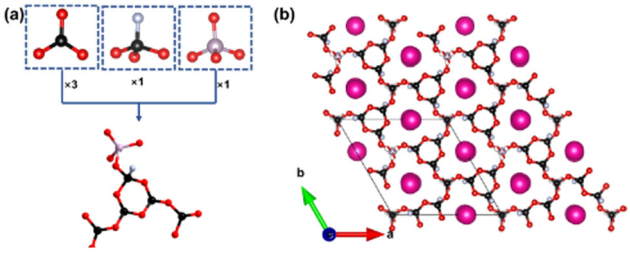

Fig. 12 (a) Anionic units of $[B_5PO_{14}F]$ FBB; (b) the crystal structure of $Rb_3B_{11}PO_{19}F_3$.

connected by B–O–B or B–O–P to form $[B_5PO_{14}F]$ FBBs, FBBs are further bonded by sharing O atoms to form 2D infinite $[B_{11}PO_{19}F_3]_\infty$ layers and these layers are attached by the B–O–P bonding to construct a 3D framework, in which cations fill between layers to balance the charge (Fig. 12). The experimental results indicate that SHG responses for $M_3B_{11}PO_{19}F_3$ (M = NH_4 and Rb) are 1.2 and 1.3 × KDP, respectively. The two crystals all have deep-UV cutoff edges (183 nm and below 200 nm). The calculated birefringence values are 0.088 and 0.080@1064 nm. The enhanced birefringence of $M_3B_{11}PO_{19}F_3$ (M = NH_4 and Rb) may be attributed to the increased polarization in the $[B_{11}PO_{19}F_3]_\infty$ layers. The shortest phase-matching wavelengths for $M_3B_{11}PO_{19}F_3$ (M = NH_4 and Rb) are 190 and 180 nm, respectively. These properties indicate that $M_3B_{11}PO_{19}F_3$ (M = NH_4 and Rb) are potential deep-UV NLO crystals.

$K_2B_4PO_9(OH)$[70] crystallizes in the $P2_1$ space group. In its structure, one $[BO_3]$, one $[BO_2(OH)]$, two $[BO_4]$ and one $[PO_4]$ units are connected by B–O–B or B–O–P to form $[B_4PO_{12}(OH)]$ FBBs, and FBBs are further bonded by sharing O atoms to form 1D infinite $[B_4PO_9(OH)]_\infty$ chains, which are separated by K cations (Fig. 13). It has a relatively large birefringence of 0.055 at 1064 nm and a short UV cutoff edge below 200 nm. However, the polycrystalline sample of $K_2B_4PO_9(OH)$ exists as an unknown phase, and no obvious SHG signal was observed. The shortest phase-matching wavelength for $K_2B_4PO_9(OH)$ is 224 nm.

5. Conclusion

In this review, we reviewed several recently synthesized borate-based short-wavelength NLO crystals with boron–oxygen skeleton modification. In the past few decades, boron-based short-wavelength NLO crystals have developed rapidly and a series of UV and even deep-UV NLO crystals with excellent properties have been discovered. In recent years, more and more borate-based NLO crystals with outstanding properties have been emerging through the modification strategy of the boron–oxygen skeleton. Research interest in new NLO crystals has stimulated the rapid development of borate derivatives, including fluorooxoborates, hydroxyborates, hydroxyfluorooxyborates, borophosphates, borosilicates and borosulfates. The boron–oxygen skeleton is modified by $[BO_xF_{4-x}]$ (x = 0–3), $[BO_x(OH)_{3-x}]$ (x = 0–2), $[BO_2(OH)_2]$, $[B(OH)_4]$, $[PO_4]$, $[SiO_4]$ and $[SO_4]$ units to a rich diversity of structures and compositions to achieve a balance between key parameters (band gap, SHG response, and birefringence). Therefore, borate-based crystals deserve further investigation as potential UV nonlinear optical crystals. The SHG response, band gap, birefringence, structure–property relationship and phase matching capabilities of the mentioned crystals were summarized in detail.

In general, the NLO crystals mentioned in this review illustrate the potential of boron–oxygen skeleton modification in obtaining UV/deep-UV NLO borate crystals. In order to effectively design the next generation of borate-based NLO materials that achieve fourth-harmonic generation in the future, many problems and challenges still exist. Based on this, we proposed some urgent challenges and prospects:

(1) The introduction of F/hydroxyl groups in the boron–oxygen backbone can fully/partially eliminate the hanging bonds of the terminal O atoms, which will help to generate a larger band gap, while the boron–oxygen anion skeleton will optimize the arrangement of the anion groups to some extent due to the presence of hydrogen bonding interactions by the introduction of hydroxyl groups, which are influenced by the directionality and saturation of the hydrogen bonds. [BOH]/[BOF] is also an important nonlinear active gene for the design of short-wavelength NLO crystals, which has a non-negligible role in the gain of SHG responses and birefringence properties. From the perspective of experimental synthesis, fluorinated and hydroxylated borates are usually thermally unstable, which also poses a challenge for the exploration of new fluorinated and hydroxylated borate crystals.

(2) Incorporation of P or S atoms into the boron–oxygen skeleton to form B–O–S or B–O–P linked borophosphates or borosulfates facilitates UV/deep-UV transmission by the introduction of non-π-conjugated units $[SO_4]$ or $[PO_4]$ and enhances linear/nonlinear optical properties by modulating the arrangement of the anionic functional group elements.

(3) The introduction of both F/hydroxyl and P/S atoms in the boron–oxygen skeleton further enriches the structural chemistry of borates and improves the optical properties, which is important for the exploration of short-wavelength NLO crystals; however, it remains a severe challenge to experimentally realize the introduction of multiple functional units. Despite the existence of centimeter-scale single crystals in some crystals, the growth of large-size single crystals still faces a great challenge. Various issues such as high melt viscosity, high volatility, phase instability and phase transition need to be considered during crystal growth. It is important to optimize crystal growth conditions

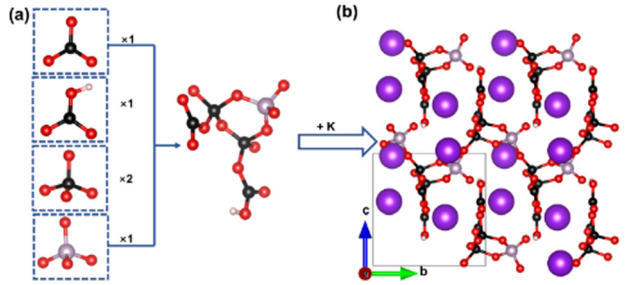

Fig. 13 (a) Anionic units of the $[B_4PO_{12}(OH)]$ FBB; (b) the crystal structure of $K_2B_4PO_9(OH)$.

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and to develop new crystal growth equipment with high precision controllers to meet the requirements under special conditions.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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